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Research Article

Total Dominator Coloring of the Hypercube Q_{2^n}

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Abstract

A proper coloring of the vertices of a graph G is called a total dominator coloring if every vertex of G is adjacent to all vertices of at least one color class. The total dominator chromatic number of G, denoted by $\chi_d^t(G)$, is the minimum number of color classes required in such a coloring. In this paper, the total dominator chromatic number of the hypercube Q_{2^n} is determined for $n \geq 3$. Also, algorithms are developed to obtain the maximum number 2^n of disjoint total dominating sets in Q_{2^n} for $n = 3$ and $n = 4$. A notable feature of this study is the introduction and utilization of novel combinatorial structures termed semi-Latin squares, pairs of mate semi-Latin squares, and chess Latin squares, which play a crucial role in constructing the corresponding colorings and domination partitions.

Keywords: total dominator coloring; total domination number; total domatic number; k-dominating colorable graph; hypercube; (pair of mate) semi-Latin square; chess Latin square.

2020 Mathematics Subject Classification: 05C15, 05C69, 05B15.

1. Introduction

Domination in graphs has been extensively studied in graph theory, and the literature on this topic has been surveyed and detailed in several books. The exact values of the domination number $\gamma(Q_n)$ of the n-dimensional hypercube Q_n, which are known only for $n \leq 9$, are given in Table 1.1.

n	1	2	3	4	5	6	7	8	9	10
$\gamma(Q_n)$	1	2	2	4	7	12	16	32	62	107-120

Table 1.1: Domination numbers of hypercubes up to dimension 10.

Also the domination numbers of Q_{2^n-1} and Q_{2^n} are given in [4] .

Theorem 1.1 (see [4]). *For any integer $n \geq 1$,*

$$\gamma(Q_{2^n-1}) = 2^{2^n-n-1} \quad \textit{and} \quad \gamma(Q_{2^n}) = 2^{2^n-n}.$$

Recently, in [2], the total domination number of the Cartesian product of a bipartite graph with the complete graph K_2 was expressed in terms of the domination number of the graph.

Theorem 1.2 (see [2]). *For any bipartite graph G,*

$$\gamma_t(G\Box K_2) = 2\gamma(G).$$

Since Q_n can be represented as the nth power of K_2 with respect to the Cartesian product operation $\Box$, that is, $Q_1 = K_2$ and $Q_n = Q_{n-1}\Box K_2$ for $n \geq 2$, the following result follows directly from Theorems 1.1 and 1.2.

Theorem 1.3 (see [2]). *For any positive integer n,*

$$\gamma_t(Q_{2^n}) = 2^{2^n-n} \quad \textit{and} \quad \gamma_t(Q_{2^n+1}) = 2^{2^n-n+1}.$$

For a given graph property $\mathcal{P}$, it is natural to consider partitions of the vertex set $V = \{V_1, V_2, \ldots, V_k\}$ such that each subset V_i $(1 \leq i \leq k)$ induces a $\mathcal{P}$-set. Such partitions are referred to as $\mathcal{P}$-colorings. When $\mathcal{P}$ denotes the property of total domination, the corresponding notion is the *total domatic number*, introduced by Cockayne, Dawes, and Hedetniemi [3].

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Motivated by the relationship between coloring and total domination, the concept of the *total dominator chromatic number* χ_d^t of a graph was first introduced in an unpublished manuscript by Hedetniemi, Hedetniemi, McRae, Rall, and Hedetniemi [5]. The first published works on this topic appear to be a 2012 paper by Vijayalekshmi and the 2015 paper by Kazemi [9]. The reader is referred to the book [5] (Part I, Section 4) and to the papers [1, 6–12] for further details on this concept.

In this paper, we proceed as follows. In Section 2, the concepts used throughout the paper are introduced. In Sections 3 and 4, by using new concepts, namely the *semi-Latin square*, *pair of mate semi-Latin squares*, and *chess Latin square*, we give some algorithms to find 2^n disjoint total dominating sets of Q_{2^n} when n is, respectively, 3 and 4. In Section 5, we show that the upper bound

$$\chi_d^t(G) \leq \gamma_t(G) + \chi(G) \quad \text{for any graph } G \text{ with } \delta(G) \geq 1, \tag{1}$$

which was given in [9], is achieved for the hypercubes Q_{2^n} for all $n \geq 3$. Finally, we conclude our paper with a question and a conjecture for future research.

2. Preliminaries

In a graph $G = (V(G), E(G))$, denote by $deg_G(v)$, $N_G(v)$, and $N_G[v]$, respectively, the *degree*, *open neighborhood*, and *closed neighborhood* of a vertex $v \in V(G)$. The *minimum degree*, *maximum degree*, and *independence number* of G are denoted by $\delta = \delta(G)$, $\Delta = \Delta(G)$, and $\alpha = \alpha(G)$, respectively. Also, a complete graph of order n is denoted by K_n, and a subgraph of G *induced* by a subset $S \subseteq V$ is denoted by $G[S]$. For any positive integers k and ℓ, we use $[\ell]$ and $k + [\ell]$ to denote the sets $\{1, 2, \ldots, \ell\}$ and $\{k+1, k+2, \ldots, k+\ell\}$, respectively. Also, for any two sets A and B, AB denotes the set $\{ab \mid a \in A,\ b \in B\}$, and $aB = AB$ where $A = \{a\}$.

For any two graphs G and H, the *Cartesian product* $G\Box H$ is a graph with vertex set $V(G) \times V(H)$ and edge set

$$E(G\Box H) = \{(g, h_1)(g, h_2) \mid h_1h_2 \in E(H)\} \cup \{(g_1, h)(g_2, h) \mid g_1g_2 \in E(G)\}.$$

Thus, $G\Box H$ is obtained from G by substituting a copy ${}_aH$ of H for every vertex a of G, and then joining the the vertex (a, c) of ${}_aH$ to the vertex (b, c) of ${}_bH$ whenever $ab \in E(G)$. By assumption, $Q_1 = K_2$, and by induction, we define $Q_n = Q_{n-1}\Box K_2$ as the *n-dimensional hypercube*. In fact, Q_n is a simple bipartite graph whose vertex set is

$$\{a_1a_2 \ldots a_n \mid a_i \in \{0, 1\} \text{ for all } i\},$$

and whose edges are the pairs of n-tuples that differ in exactly one position.

A *dominating set* S of a graph G is a subset of $V(G)$ such that every vertex in $V(G) - S$ is adjacent to at least one vertex in S. A *total dominating set* (briefly, TD-set) S of G, where G has no isolated vertex, is a subset of $V(G)$ such that every vertex in $V(G)$ is adjacent to at least one vertex in S. The *domination number* $\gamma(G)$ and the *total domination number* $\gamma_t(G)$ of G are the minimum cardinalities of a dominating set and a TD-set of G, respectively. The *total domatic number* $tdom(G)$ of G is the maximum order of a vertex partition of G into total dominating sets.

For the set $[k]$ of k colors, a function $f : V(G) \to [k]$ is a *proper k-coloring* of G if f is a homomorphism from G to the complete graph K_k with vertex set $[k]$; that is, $f(x) \neq f(y)$ for every edge $xy \in E(G)$. A graph that has a proper k-coloring is called k-colorable. The *chromatic number* $\chi(G)$ of a graph G is the minimum number k of colors needed in a proper coloring of G.

In a proper coloring of a graph, a *color class* is a set consisting of all vertices assigned the same color. If f is a proper coloring of G with color classes $V_1, V_2, \ldots, V_\ell$ such that every vertex in V_i has color i, we simply write $f = (V_1, V_2, \ldots, V_\ell)$. For a proper coloring $f = (V_1, V_2, \ldots, V_\ell)$ of a graph G, a vertex v is called a *common neighbor* of V_i, or equivalently, we say that v is *totally dominated by* V_i, or that V_i *totally dominates* v, and we write $v \succ_t V_i$ if $V_i \subseteq N_G(v)$. Otherwise, we write $v \not\succ_t V_i$.

A vertex v is called a *private neighbor* of V_i with respect to f if $v \succ_t V_i$ and $v \not\succ_t V_j$ for all $j \neq i$. The set of all common neighbors of V_i, denoted by $CN_G(V_i)$ or simply by $CN(V_i)$, is called the *common neighborhood* of V_i in G with respect to f.

A new type of vertex coloring of a graph was introduced in [9] as follows.

Definition 2.1. A *total dominator coloring,* briefly TD-coloring, of a graph G without isolated vertices is a proper coloring of G such that every vertex of G is adjacent to all vertices of some color class. The minimum number of color classes in a TD-coloring of G is called the *total dominator chromatic number* of G, denoted by $\chi_d^t(G)$. A TD-coloring of G using exactly $\chi_d^t(G)$ colors is called a *minimum total dominator coloring* (or simply, a *min-TD-coloring*.

3. The Hypercube Q_8

Here, we present eight disjoint total dominating sets of Q_8 with minimum cardinality. First, we give a new definition.

Definition 3.1. For a k-regular graph G, let $S \subseteq V(G)$ be a set such that the induced subgraph $G[S]$ is k-colorable. The graph G is called *k-dominating colorable* with the *k-dominating coloring set* S if every vertex of $V(G) - S$ is adjacent to k vertices of S having k different colors.

If it is not ambiguous from the context, we will simply use the terms *dominating colorable* and *dominating coloring set*. As shown in Figure 3.1, the hypergraph Q_4 is dominating colorable.

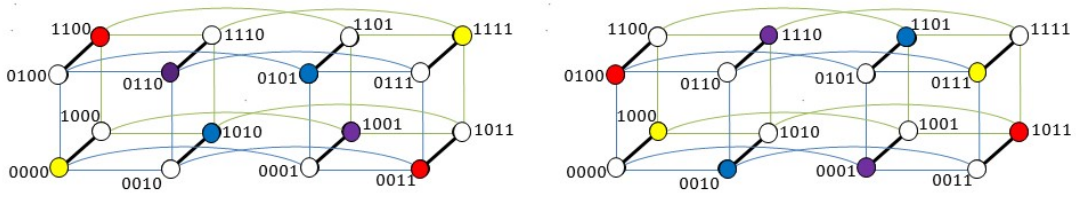


Figure 3.1: Two disjoint dominating coloring sets of Q_4.

Lemma 3.1. $tdom(Q_4) = 4$.

Proof. Clearly, the four sets

$$Y_1 = \{100, 011\}, \quad Y_2 = \{010, 101\}, \quad Y_3 = \{110, 001\}, \quad Y_4 = \{000, 111\}$$

are disjoint minimum dominating sets of Q_3. As shown in Figure 3.2, since the sets $X_i = \{0, 1\}Y_i$, $1 \leq i \leq 4$, are four disjoint minimum total dominating sets of Q_4, we conclude that the total domatic number of Q_4 is 4. □

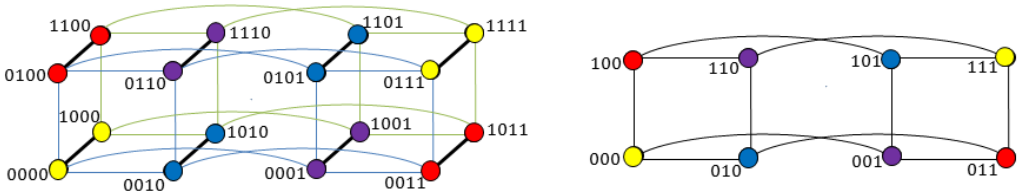


Figure 3.2: The four disjoint minimum total dominating sets X_i of Q_4 (left) and the four disjoint minimum dominating sets Y_i of Q_3 (right).

In Figure 3.3, for simplicity, we use (a) or (b) to represent the hypercube Q_4, and (a) or (c) to denote the hypercube Q_n as the Cartesian product $Q_4 \square H$, where $H = Q_{n-4}$.

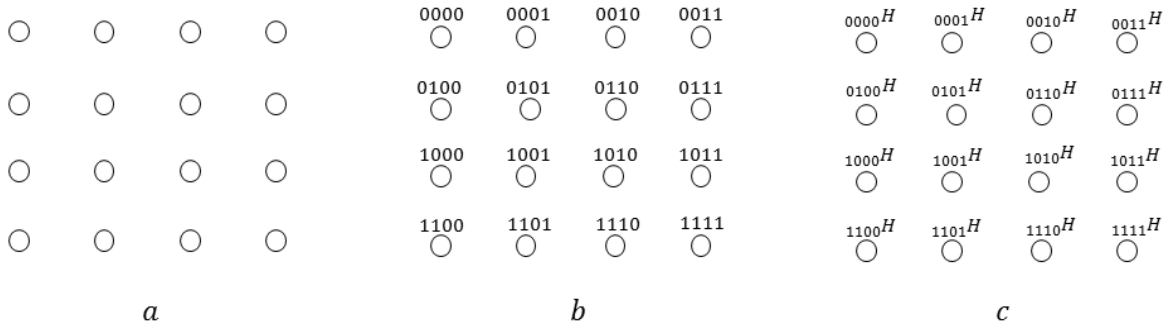


Figure 3.3: (a) and (b): the hypercube Q_4; and (a) and (c): the Cartesian product $Q_4 \square H$.

For example, in Figure 3.4, two disjoint dominating coloring sets of Q_4 shown in Figure 3.1 and the four disjoint min-TD-sets X_i of Q_4 shown in Figure 3.2 are illustrated using a new representation in Figure 3.4.

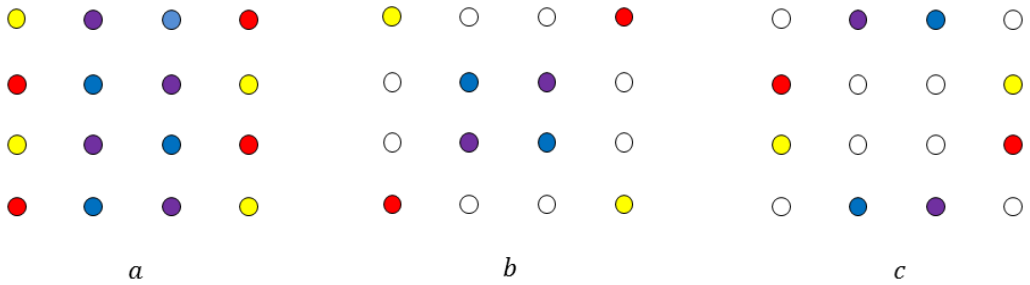


Figure 3.4: (a): The four disjoint min-TD-sets X_i of Q_4 shown in Figure 3.1; (b) and (c): The two disjoint dominating coloring sets of Q_4 shown in Figure 3.2.

In the following two subsections, we prove that the total domatic number of Q_8 is eight.

An Algorithm to Construct Eight Disjoint Min-TD-Sets of Q_8

- Step 1. Consider Figure 3.4(b), which shows an independent set of the bipartite graph Q_4. Using the facts that $Y_1 \cup Y_2$ and $Y_3 \cup Y_4$ are two disjoint min-TD-sets of Q_3, and that the set of colored vertices forms an independent set of Q_4, we assign the pairs X_1 and X_2 to the red and yellow cells, and the pairs X_3 and X_4 to the blue and purple cells, respectively. Accordingly, we obtain the set

$$\mathcal{D}_1 = \{0000, 1111\}X_1 \cup \{0011, 1100\}X_2 \cup \{0101, 1010\}X_3 \cup \{0110, 1001\}X_4.$$

- Step 2. Apply the permutation $\sigma = (1\ 2)(3\ 4)$ to the indices of the sets X_i ($1 \leq i \leq 4$), and obtain

$$\mathcal{D}_2 = \{0000, 1111\}X_2 \cup \{0011, 1100\}X_1 \cup \{0101, 1010\}X_4 \cup \{0110, 1001\}X_3.$$

- Step 3. Apply the permutation $\delta = (1\ 3)(2\ 4)$ to the indices of the sets X_i ($1 \leq i \leq 4$), and obtain the sets $D_{\delta(i)}$ from $\mathcal{D}_i$ for $i = 1, 2$. Thus,

$$\mathcal{D}_3 = \{0000, 1111\}X_3 \cup \{0011, 1100\}X_4 \cup \{0101, 1010\}X_1 \cup \{0110, 1001\}X_2,$$

$$\mathcal{D}_4 = \{0000, 1111\}X_4 \cup \{0011, 1100\}X_3 \cup \{0101, 1010\}X_2 \cup \{0110, 1001\}X_1.$$

- Step 4. Consider the second independent set of the bipartite graph Q_4 shown in Figure 3.4(c), and repeat the above procedure to obtain the following four sets

$$\mathcal{D}_5 = \{0000, 1111\}X_1 \cup \{0011, 1100\}X_2 \cup \{0101, 1010\}X_3 \cup \{0110, 1001\}X_4,$$

$$\mathcal{D}_6 = \{0000, 1111\}X_2 \cup \{0011, 1100\}X_1 \cup \{0101, 1010\}X_4 \cup \{0110, 1001\}X_3,$$

$$\mathcal{D}_7 = \{0000, 1111\}X_3 \cup \{0011, 1100\}X_4 \cup \{0101, 1010\}X_1 \cup \{0110, 1001\}X_2,$$

$$\mathcal{D}_8 = \{0000, 1111\}X_4 \cup \{0011, 1100\}X_3 \cup \{0101, 1010\}X_2 \cup \{0110, 1001\}X_1,$$

which are shown in two forms in Figures 3.5 and 3.6.

Two Claims

- Claim 1. For each $1 \leq i \leq 8$, $\mathcal{D}_i$ is a min-TD-set of Q_8.

We know that $Q_8 = Q_4 \square Q_4$, and each of the sets X_i, $1 \leq i \leq 4$, is a min-TD-set of Q_4 by Lemma 3.1. Let

$$\mathcal{D}_i = \{\{a_i, \overline{a_i}\}X_i \mid 1 \leq i \leq 4\}$$

for some $a_i = b_1b_2b_3b_4 \in V(Q_4)$ and $\overline{a_i} = \overline{b_1}\,\overline{b_2}\,\overline{b_3}\,\overline{b_4}$ (notice that $\overline{x} = 0$ if and only if $x = 1$). Since every X_i is a TD-set of Q_4, every vertex of $\{a_i, \overline{a_i}\}Q_4$ is dominated by a vertex of $\{a_i, \overline{a_i}\}X_i$, and hence by $\mathcal{D}_i$. Let ${}_aQ_4$ be a copy of Q_4 in which

$$a \notin \{a_i, \overline{a_i} \mid 1 \leq i \leq 4\}.$$

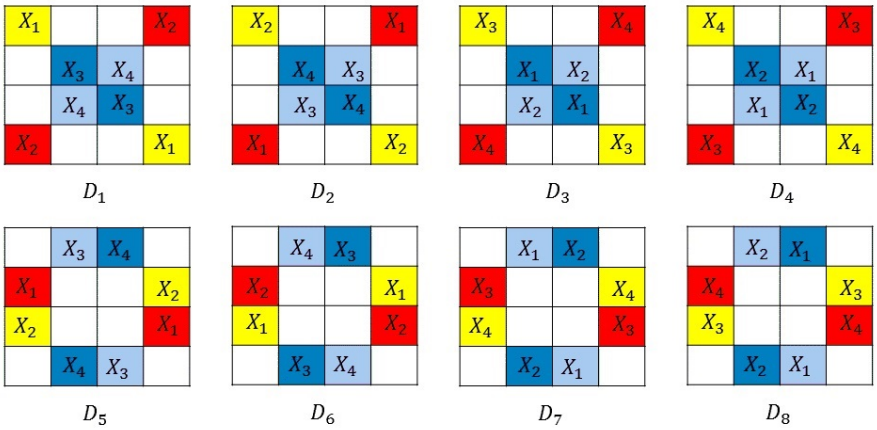


Figure 3.5: The sets D_i for $1 \leq i \leq 8$.

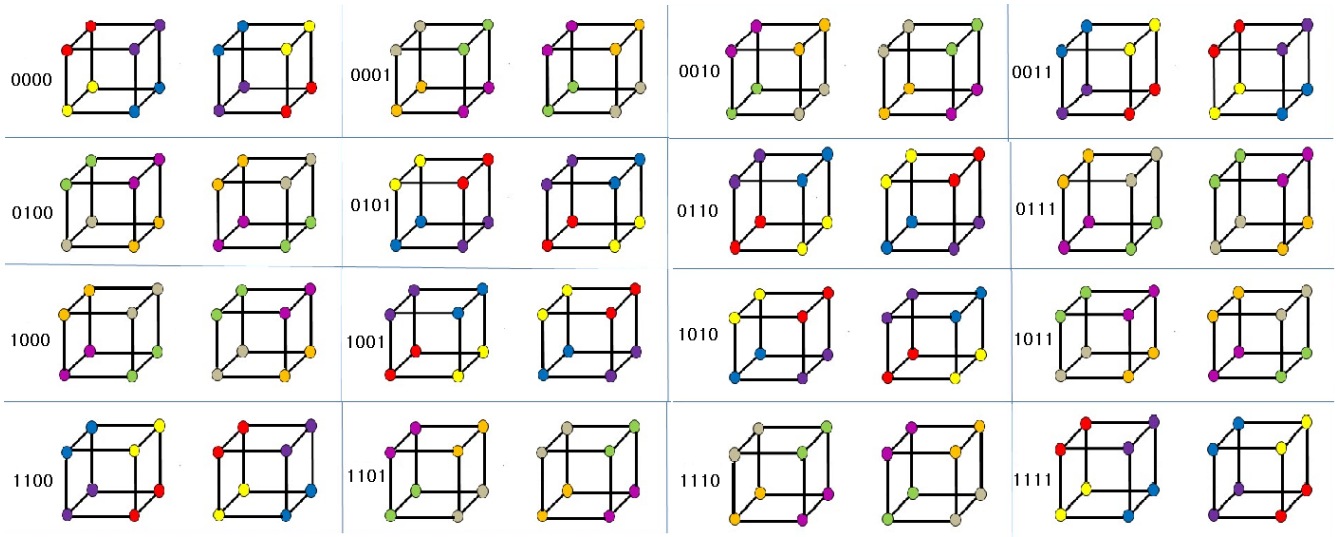


Figure 3.6: The sets $\mathcal{D}_i$ for $1 \leq i \leq 8$ (each set of vertices with the same color represents a $\mathcal{D}_i$).

By the facts that $\{a_i, \overline{a_i} \mid 1 \leq i \leq 4\}$ is a 4-dominating coloring of Q_4 and the sets X_1, X_2, X_3, X_4 are four disjoint TD-sets of Q_4, we see that for some

$$\{b_i \mid 1 \leq i \leq 4\} \subseteq \{a_i, \overline{a_i} \mid 1 \leq i \leq 4\},$$

the neighbors of $b_1X_1 \cup b_2X_2 \cup b_3X_3 \cup b_4X_4$ in ${}_aQ_4$ make the partition

$$V({}_aQ_4) = aX_1 \cup aX_2 \cup aX_3 \cup aX_4,$$

and this guarantees that every vertex of ${}_aQ_4$ is dominated by only one vertex of

$$b_1X_1 \cup b_2X_2 \cup b_3X_3 \cup b_4X_4,$$

and hence by only one vertex of $\mathcal{D}_i$. Hence, for each $1 \leq i \leq 8$, $\mathcal{D}_i$ is a TD-set of Q_8, and also it is minimum since $\gamma_t(Q_8) = 2^5 = |\mathcal{D}_i|$.

- Claim 2. For each $1 \leq i < j \leq 8$, $\mathcal{D}_i \cap \mathcal{D}_j = \emptyset$.

  Since the sets $\mathcal{D}_1, \ldots, \mathcal{D}_4$ and $\mathcal{D}_5, \ldots, \mathcal{D}_8$ are made by using the two disjoint 4-dominating coloring sets

  $$\mathcal{C}_1 = \{0000, 0011, 0101, 0110, 1001, 1010, 1100, 1111\} \quad \text{and} \quad \mathcal{C}_2 = \{0001, 0010, 0100, 0111, 1000, 1011, 1101, 1110\},$$

  respectively, we have $\mathcal{D}_i \cap \mathcal{D}_j = \emptyset$ when $1 \leq i \leq 4 < j \leq 8$. So, we may assume $1 \leq i < j \leq 4$ or $5 \leq i < j \leq 8$. Then, since for any copy aX_p of X_p in $\mathcal{D}_i$ and any copy aX_q of X_q in $\mathcal{D}_j$,

  $$aX_p \cap aX_q = a(X_p \cap X_q) = a\emptyset = \emptyset,$$

  again we have $\mathcal{D}_i \cap \mathcal{D}_j = \emptyset$.

4. The Hypercube Q_{16}

To prove that the total domatic number of the hypercube Q_{16} is 16, we introduce and utilize novel combinatorial structures termed *semi-Latin squares*, *pairs of mate semi-Latin squares*, and *chess Latin squares*, which play a crucial role in constructing the corresponding colorings and domination partitions.

The Semi-Latin Square, Pair of Mate Semi-Latin Square, and Chess Latin Square

Let $V(Q_4) = X \cup Y$ be a partition of the vertex set of the hypercube Q_4 into the independent sets X and Y. Assume that M is a 4×4 chessboard in which each cell corresponds to a vertex of Q_4, as illustrated in Figure 3.3. Let $\mathbb{X}$ denote the 4×4 chessboard obtained from M by coloring the cells corresponding to the vertices of X *blue*, and the remaining cells *white*. Similarly, define the 4×4 chessboard $\mathbb{Y}$ corresponding to Y. We now construct the following 16×16 chessboard.

$$L = \begin{pmatrix} \mathbb{X} & \mathbb{Y} & \mathbb{Y} & \mathbb{X} \\ \mathbb{Y} & \mathbb{X} & \mathbb{X} & \mathbb{Y} \\ \mathbb{Y} & \mathbb{X} & \mathbb{X} & \mathbb{Y} \\ \mathbb{X} & \mathbb{Y} & \mathbb{Y} & \mathbb{X} \end{pmatrix}.$$

If we can fill the white cells of L with the numbers $1, 2, \ldots, 8$ such that each of them appears in every row, every column, and each of the subchessboards $\mathbb{X}$ and $\mathbb{Y}$, then we call L a *semi-Latin square* of order 8. Notice that, by permuting the four columns of L according to $(1\ 2)(3\ 4)$, we obtain the *complement* of L,

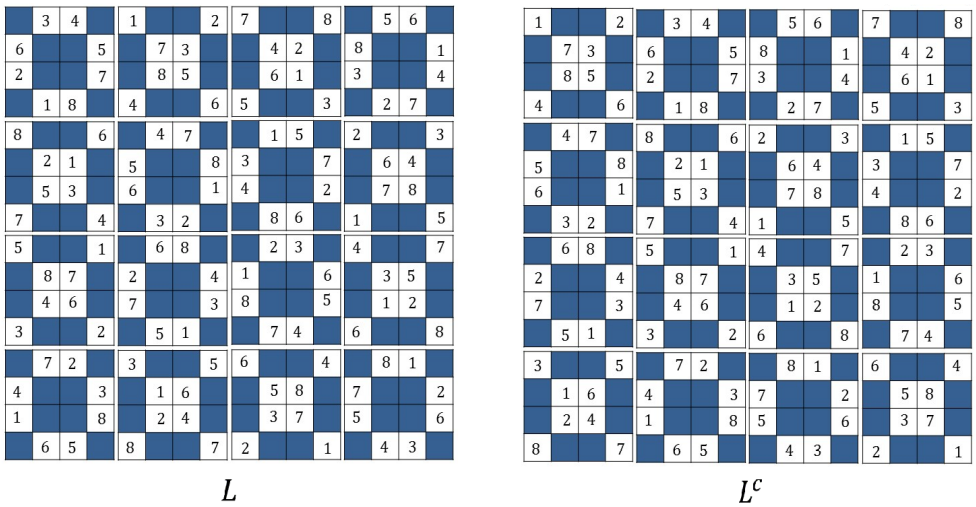
$$L^c = \begin{pmatrix} \mathbb{Y} & \mathbb{X} & \mathbb{X} & \mathbb{Y} \\ \mathbb{X} & \mathbb{Y} & \mathbb{Y} & \mathbb{X} \\ \mathbb{X} & \mathbb{Y} & \mathbb{Y} & \mathbb{X} \\ \mathbb{Y} & \mathbb{X} & \mathbb{X} & \mathbb{Y} \end{pmatrix},$$

as another semi-Latin square of order 8 obtained from L. We call L and L^c a *pair of mate semi-Latin squares* of order 8. Figure 4.1 shows a pair of mate semi-Latin squares of order 8.

If we add 8 to all of the integers in L^c and then display the pair of mate semi-Latin squares L and L^c on the same chessboard, we obtain a Latin square of order 16 on the set $[16]$, with the additional property that all of the numbers of $[8]$ appear in the white cells of every subchessboard $\mathbb{X}$, and all of the numbers of $8 + [8]$ appear in the blue cells of every subchessboard $\mathbb{Y}$. We call this Latin square the *chess Latin square* $CLS(L, L^c)$ of order 16. The Latin square of order 16 shown in Figure 4.2 is the chess Latin square corresponding to the pair of mate semi-Latin squares L and L^c in Figure 4.1.

Since, by the inverse process, we can also obtain a pair of mate semi-Latin squares of order 8 from a chess Latin square of order 16, we have the following theorem.

Theorem 4.1. *There exists a chess Latin square of order 16 if and only if there exists a pair of mate semi-Latin squares of order 8.*



L L^c

Figure 4.1: A pair of mate semi-Latin squares L and L^c of order 8.

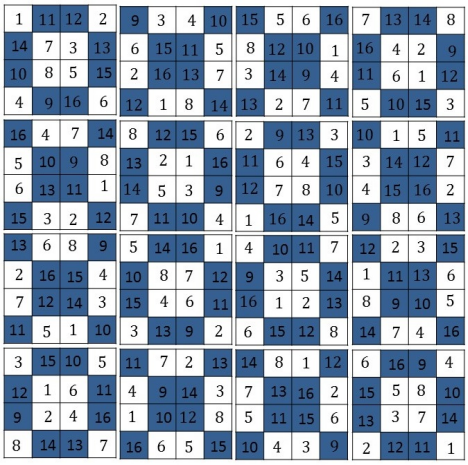


Figure 4.2: The chess Latin square $CLS(L, L^c)$ of order 16.

Also, if we consider a semi-Latin square L of order 8 and define

$$V_i = \{\, v \in V(Q_8) \mid i \text{ appears in the cell corresponding to } v \text{ in } L \,\},$$

then we can state that Q_8 is 8-dominating colorable with the 8-dominating coloring set

$$T = V_1 \cup V_2 \cup \cdots \cup V_8.$$

In fact, T is the set of vertices corresponding to all of the blue (or white) cells. Hence, every semi-Latin square of order 8 is equivalent to an 8-dominating coloring of Q_8, and therefore the next theorem follows from Theorem 4.1.

Theorem 4.2. *Let $X \cup Y$ be the partition of the vertex set of the hypercube Q_8 into the independent sets X and Y. Then the following statements are equivalent:*

- *There exists a chess Latin square of order 16.*
- *There exists a pair of mate semi-Latin squares of order 8.*
- *The hypercube Q_8 is 8-dominating colorable with two disjoint 8-dominating coloring sets X and Y.*

Total Domatic Number of Q_{16}

Proposition 4.1. $tdom(Q_{16}) = 16$.

Proof. We know from the previous section that the sets $\mathcal{D}_1, \mathcal{D}_2, \ldots, \mathcal{D}_8$ are eight disjoint min-TD-sets of Q_8. Consider the semi-Latin square L in Figure 4.1. Since $Q_{16} = Q_8 \square Q_8$ and every semi-Latin square corresponds to Q_8, if we place a copy of Q_8 in each cell of L and use the notation given in Figure 3.4, then each of the pair of mate semi-Latin squares L and L^c in Figure 4.1 represents the hypercube Q_{16}. By choosing $\mathcal{D}_i$ in the copy of Q_8 whose corresponding cell contains i, we obtain the set

$$\mathcal{L} = \bigcup_{i=1}^{8} \mathcal{A}_i \mathcal{D}_i,$$

where

$$\begin{aligned}
\mathcal{A}_1 &= \{00001101,\ 00010000,\ 00101010,\ 00110111,\ 01000101,\ 01011011,\ 01100001,\ 01111100\} \\
&\cup \{11110010,\ 11101111,\ 11010101,\ 11001000,\ 10111010,\ 10100100,\ 10011110,\ 10000011\}, \\
\mathcal{A}_2 &= \{00001000,\ 00010011,\ 00100110,\ 00111101,\ 01000101,\ 01011110,\ 01101011,\ 01110000\} \\
&\cup \{11110111,\ 11101100,\ 11011001,\ 11000010,\ 10111010,\ 10100001,\ 10010100,\ 10001111\}, \\
\mathcal{A}_3 &= \{00000001,\ 00010110,\ 00101111,\ 00111000,\ 01001010,\ 01011101,\ 01100100,\ 01110011\} \\
&\cup \{11111110,\ 11101001,\ 11010000,\ 11000111,\ 10110101,\ 10100010,\ 10011011,\ 10001100\}, \\
\mathcal{A}_4 &= \{00000010,\ 00011100,\ 00100101,\ 00111011,\ 01001111,\ 01010001,\ 01101000,\ 01110110\} \\
&\cup \{11111101,\ 11100011,\ 11011010,\ 11000100,\ 10110000,\ 10101110,\ 10010111,\ 10001001\}, \\
\mathcal{A}_5 &= \{00000111,\ 00011010,\ 00101100,\ 00110001,\ 01001001,\ 01010000,\ 01100010,\ 01111111\} \\
&\cup \{11111000,\ 11100101,\ 11011001,\ 11001110,\ 10110110,\ 10101111,\ 10011101,\ 10000000\},
\end{aligned}$$

$$\begin{aligned}
\mathcal{A}_6 &= \{00000100,\ 00011111,\ 00101001,\ 00110010,\ 01000011,\ 01011000,\ 01101110,\ 01110101\}\\
&\cup \{11111011,\ 11100000,\ 11010110,\ 11001101,\ 10111100,\ 10100111,\ 10010001,\ 10001010\},\\
\mathcal{A}_7 &= \{00001011,\ 00010101,\ 00100000,\ 00111110,\ 01001100,\ 01010010,\ 01100111,\ 01111001\}\\
&\cup \{11110100,\ 11101010,\ 11011111,\ 11000001,\ 10110011,\ 10101101,\ 10011000,\ 10000110\},\\
\mathcal{A}_8 &= \{00001110,\ 00011001,\ 00100011,\ 00110100,\ 01000000,\ 01010111,\ 01101101,\ 01111010\}\\
&\cup \{11110001,\ 11100110,\ 11011100,\ 11001011,\ 10111111,\ 10101000,\ 10010010,\ 10000101\}.
\end{aligned}$$

In a similar way, by considering L^c, we may obtain the set

$$\mathcal{L}^c = \bigcup_{i=1}^{8} \mathcal{B}_i\mathcal{D}_i$$

of Q_{16}, which is disjoint from $\mathcal{L}$, where

$$\begin{aligned}
\mathcal{B}_1 &= \{00000000,\ 00011101,\ 00100111,\ 00111010,\ 01001011,\ 01010110,\ 01101100,\ 01110001\}\\
&\cup \{11111111,\ 11100010,\ 11011000,\ 11000101,\ 10110100,\ 10101001,\ 10010011,\ 10001110\},\\
\mathcal{B}_2 &= \{00000011,\ 00011000,\ 00101101,\ 00110110,\ 01001110,\ 01010101,\ 01100000,\ 01111011\}\\
&\cup \{11111100,\ 11100111,\ 11010010,\ 11001001,\ 10110001,\ 10101010,\ 10011111,\ 10000100\},\\
\mathcal{B}_3 &= \{00000110,\ 00010001,\ 00101000,\ 00111111,\ 01001101,\ 01011010,\ 01100011,\ 01111111\}\\
&\cup \{11111001,\ 11101110,\ 11010111,\ 11000000,\ 10110010,\ 10100101,\ 10011100,\ 10000000\},\\
\mathcal{B}_4 &= \{00001100,\ 00010010,\ 00101011,\ 00110101,\ 01000001,\ 01011111,\ 01100110,\ 01111000\}\\
&\cup \{11110011,\ 11101101,\ 11010100,\ 11001010,\ 10111110,\ 10100000,\ 10011001,\ 10000111\},\\
\mathcal{B}_5 &= \{00001010,\ 00010111,\ 00100001,\ 00111100,\ 01000100,\ 01011001,\ 01101111,\ 01110010\}\\
&\cup \{11110101,\ 11101000,\ 11011110,\ 11000011,\ 10111011,\ 10100110,\ 10010000,\ 10001101\},\\
\mathcal{B}_6 &= \{00001111,\ 00010100,\ 00100010,\ 00111001,\ 01001000,\ 01010011,\ 01100101,\ 01111110\}\\
&\cup \{11110000,\ 11101011,\ 11011101,\ 11000110,\ 10110111,\ 10101100,\ 10010101,\ 10000001\},\\
\mathcal{B}_7 &= \{00000101,\ 00011011,\ 00101110,\ 00110000,\ 01000010,\ 01011100,\ 01101001,\ 01110111\}\\
&\cup \{11111010,\ 11100100,\ 11010001,\ 11001111,\ 10111101,\ 10100011,\ 10010110,\ 10001000\},\\
\mathcal{B}_8 &= \{00001001,\ 00011110,\ 00100100,\ 00110011,\ 01000111,\ 01010000,\ 01101010,\ 01111101\}\\
&\cup \{11110110,\ 11100001,\ 11011011,\ 11001100,\ 10111000,\ 10101111,\ 10010101,\ 10000010\}.
\end{aligned}$$

Since every $\mathcal{D}_i$ is a min-TD-set of Q_8, every vertex of $\mathcal{A}_iQ_8$ is dominated by exactly one vertex of $\mathcal{A}_i\mathcal{D}_i$, and hence by exactly one vertex of $\mathcal{L}$. Let ${}_aQ_8$ denote a copy of Q_8 where $a \notin \bigcup_{i=1}^{8} \mathcal{A}_i$. By Theorem 4.2, since the set of white cells with their numbers forms an 8-dominating coloring of Q_8, and the sets $\mathcal{D}_1, \ldots, \mathcal{D}_8$ are eight disjoint min-TD-sets of Q_8, we conclude that for some $\{b_i \mid 1 \leq i \leq 8\} \subseteq \bigcup_{i=1}^{8} \mathcal{A}_i$, the set of neighbors of $b_1\mathcal{D}_1 \cup b_2\mathcal{D}_2 \cup \cdots \cup b_8\mathcal{D}_8$ in ${}_aQ_8$ produces the partition

$$V({}_aQ_8) = a\mathcal{D}_1 \cup a\mathcal{D}_2 \cup \cdots \cup a\mathcal{D}_8.$$

This guarantees that every vertex of ${}_aQ_8$ is dominated by exactly one vertex of $b_1\mathcal{D}_1 \cup b_2\mathcal{D}_2 \cup \cdots \cup b_8\mathcal{D}_8$, and hence by exactly one vertex of $\mathcal{L}$. Therefore, $\mathcal{L}$ is a TD-set of Q_{16}. Since also $\gamma_t(Q_{16}) = 2^{12} = |\mathcal{L}|$ by Theorem 1.3, it follows that $\mathcal{L}$ is a min-TD-set of Q_{16}. Similarly, $\mathcal{L}^c$ is a min-TD-set of Q_{16}.

Now, we present sixteen disjoint min-TD-sets of Q_{16}. Let σ be a permutation of order 8 on the set $[8]$. The fact that for each $1 \leq i \leq 8$, σ^i generates two other pairs of mate semi-Latin squares from L and L^c of order 16, guarantees that for every $1 \leq i \leq 8$, the sets $\sigma^i(\mathcal{L})$ and $\sigma^i(\mathcal{L}^c)$ are min-TD-sets of Q_{16}. Since X and Y are two disjoint 8-dominating coloring sets of Q_8, we have

$$\sigma^i(\mathcal{L}) \cap \sigma^j(\mathcal{L}^c) = \emptyset \quad \text{for all } 1 \leq i \neq j \leq 8,$$

by Theorem 4.2. Moreover, it is clear that

$$\sigma^i(\mathcal{L}) \cap \sigma^j(\mathcal{L}) = \emptyset \quad \text{and} \quad \sigma^i(\mathcal{L}^c) \cap \sigma^j(\mathcal{L}^c) = \emptyset \quad \text{for all } 1 \leq i < j \leq 8.$$

Hence, the collection

$$\{\sigma^i(\mathcal{L}),\ \sigma^i(\mathcal{L}^c)\ :\ 1 \leq i \leq 8\}$$

consists of sixteen disjoint min-TD-sets of Q_{16}, and therefore

$$tdom(Q_{16}) = 16.$$

□

5. Total Dominator Coloring of Q_{2^n}

The total dominator chromatic numbers of Q_n were determined in [8] for $1 \leq n \leq 4$.

Proposition 5.1 (see [8]). *For any* $1 \leq n \leq 4$,

$$\chi_d^t(Q_n) = \begin{cases} 2, & \text{if } n = 1, \\ 2^{n-1}, & \text{if } n = 2, 3, \\ 2^{n-2} + 2, & \text{if } n = 4. \end{cases}$$

Before determining the total dominator chromatic number of Q_{2^n} for $n \geq 3$, we first examine the structure of hypercubes in more detail. Recall that $\alpha(Q_n) = 2^{n-1}$.

Fact 5.1. *Let* $(V_1, V_2, \cdots, V_\ell)$ *be a minimal TD-coloring of* Q_n *and let* $a_i = |\{V_k \mid |V_k| = i\}|$ *for each* $1 \leq i \leq 2^{n-1}$. *Then the following hold.*

(1) $V(Q_n) = \bigcup_{i=1}^{\ell} CN(V_i)$ *and so* $2^n \leq \sum_{i=1}^{\ell} |CN(V_i)|$.

(2) *For any two non-adjacent vertices* x *and* y, $N(x) \cap N(y) \neq \emptyset$ *implies* $|N(x) \cap N(y)| = 2$.

Proof: Let $z \in N(x) \cap N(y)$, *and without loss of generality, we may assume* $z = 00\cdots0$. *Since only one position of* x *and* y *is 1 and these positions are not same, again we may assume* $x = 100\cdots0$ *and* $y = 010\cdots0$. *This implies* $N(x) \cap N(y) = \{z, 110\cdots0\}$.

(3) *If* $|V_i| > n$, *then* $|CN(V_i)| = 0$ (*because* $V_i \subseteq N(v)$ *when* $v \succ V_i$, *and* $|N(v)| = n$).

(4) *If* $3 \leq |V_i| \leq n$, *then* $|CN(V_i)| \leq 1$, *and* $CN(V_i) = \{v\}$ *if and only if* $V_i \subseteq N(v)$.

Proof: let $a \in N(x) \cap N(y) \cap N(z)$ *where* $x, y, z \in V_i$ *for some* i, *and without loss of generality, we may assume* $a = 00\cdots0$, $x = 100\cdots0$, $y = 010\cdots0$, $z = 0010\cdots0$. *Let* $b = b_1\cdots b_n \in N(x) \cap N(y) \cap N(z) - \{a\}$. *Since* $a, b \in N(x) \cap N(y)$ *and* $a \neq b$, *we have* $b = 110\cdots0$ *which is not adjacent to* z.

(5) *If* $|V_i| = 2$, *then* $|CN(V_i)| = 0$ *or* 2 (*by* (2), (4)). *More* $|V_i| = 2$ *if and only if* $|CN(V_i)| = 2$ *when* $|CN(V_i)| \neq 0$ *and* $n \geq 3$.

(6) *For each* $1 \leq i \leq \ell$, $|V_i| = 1$ *if and only if* $|CN(V_i)| = n$.

(7) $2^n \leq na_1 + 2a_2 + a_3 + \cdots + a_n \leq (n-1)a_1 + a_2 + \ell$ (*by* (1), (3), (4), (5), (6)).

Theorem 5.1. *For any* $n \geq 3$,

$$\chi_d^t(Q_{2^n}) = 2^{2^n - n} + 2.$$

Proof. For any $n \geq 3$, assume that $\chi_d^t(Q_{2^n}) \leq 2^{2^n-n} + 1$, and let $(V_1, V_2, \ldots, V_\ell)$ be a minimal TD-coloring of Q_{2^n} where $|V_i| \leq |V_j|$ for $1 \leq i < j \leq \ell$. Since $|V_1| \leq \cdots \leq |V_{\ell-1}| \leq 2^n \leq |V_\ell|$ by Fact 5.1 (3), we consider two cases.

Case 1. $|V_\ell| > 2^n$. Then, by Fact 5.1 (1), we have $|CN(V_i)| = 2^n$ for each $1 \leq i < \ell$, and $CN(V_i) \cap CN(V_j) = \emptyset$ whenever $i \neq j$. Hence, $|V_i| = 1$ for each $1 \leq i < \ell$ by Fact 5.1 (6), which means that the set $S = V_1 \cup V_2 \cup \cdots \cup V_{\ell-1}$ is a TD-set of Q_{2^n} and $Q_{2^n} - Q_{2^n}[S]$ is an empty graph, a contradiction.

Case 2. $|V_\ell| = 2^n$. Let $t = |\{\, i \mid |V_i| = 2^n \,\}|$. Then, by Fact 5.1 (7),

$$2^{2^n} \leq (\ell - t)(2^n - 1) + t \leq (2^{2^n-n} + 1 - t)(2^n - 1) + 2^n t,$$

which implies

$$2^{2^n-n} - (2^n - 1) \leq t \leq 2^{2^n-n} + 1,$$

that is, $|\{\, i \mid |V_i| < 2^n \,\}| \leq 2^n + 1$. Thus,

$$2^{2^n} \leq \sum_{i=1}^{\ell-t} |CN(V_i)| + \sum_{i=\ell-t+1}^{\ell} |CN(V_i)| \leq (\ell - t)2^n + t \leq (2^{2^n-n} + 1 - t)2^n + t = 2^{2^n} + 2^n - (2^n - 1)t,$$

by Fact 5.1 (1). Then $(2^n - 1)t \leq 2^n$, and so $t = 1$, which implies $n \leq 2$, a contradiction.

Therefore,

$$\begin{aligned} \chi_d^t(Q_{2^n}) &\geq 2^{2^n-n} + 2 \\ &= \gamma_t(Q_{2^n}) + \chi(Q_{2^n}) \quad \text{(by Theorem 1.3)}, \end{aligned}$$

and hence, $\chi_d^t(Q_{2^n}) = 2^{2^n-n} + 2$ for all $n \geq 3$ by (1). □

6. A Question and a Conjecture

We recall that a graph is called *well-totally-dominated* if all of its minimal total dominating sets have the same size. Consequently, its upper total domination number and total domination number are equal. This naturally leads to the following question.

Question 6.1. *Is there any well-totally-dominated hypercube?*

Since the total domination number of a graph is a lower bound for its total dominator chromatic number, by (1) and Theorem 1.3, we have

$$2^{2^n-n+1} \leq \chi_d^t(Q_{2^n+1}) \leq 2^{2^n-n+1}+2. \tag{2}$$

We conjecture that the total dominator chromatic number of the hypercube Q_{2^n+1} attains the upper bound in (2).

Conjecture 6.1. *For any integer $n \geq 2$,*

$$\chi_d^t(Q_{2^n+1}) = 2^{2^n-n+1}+2.$$

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